Abhay Mishra

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Profile

I'm an ambitious and curious Engineering Physics student at Uppsala University with a passion for computational physics, embedded systems, and automation. I graduated high school two years early, at 17, and currently completing additional courses to accelerate my university studies. I always try challenge myself both academically and technically to grow as an engineer and problem solver. I enjoy taking initiative, learning fast, and actively engaging in various student and community activities.

Education

Master of Science in Engineering – Engineering Physics — August 2024 – June 2028
Uppsala University — *Uppsala, Sweden*

High School Diploma – Teknikprogrammet — August 2021 – June 2024
NTI - Gymanasium — *Uppsala, Sweden*

Work Experience

Junior Software Engineer — September 2025 – Present
BruhnBruhn Innovation — *Uppsala, Sweden*

Student Ambassador — September 2025 - Present
Uppsala University - Engineering and Science Faculty — *Uppsala, Sweden*

SciFest Functionary — October 2025
Uppsala University - Engineering and Science Faculty — *Uppsala, Sweden*

Summer Research Intern - Division of Materials Theory — June 2025 – September 2025
Uppsala University - Engineering and Science Faculty — *Uppsala, Sweden*

Substitute Teacher – Tutoring Support — September 2024 – June 2025
NTI Gymnasium — *Uppsala, Sweden*

Study Coach — June 2024 – October 2024
Allakando — *Uppsala, Sweden*

Field Assistant and Statistical Analysis Intern — June 2024 – July 2024
SLU (Swedish University of Agricultural Sciences) — *Uppsala, Sweden*

Technical Experience & Projects

Longitudinal Spin Susceptibility Solver - RSPT Inputs 2025 — June 2025 – September 2025
Research Project – Uppsala University — *Python*

- Developed a computational physics solver for calculating longitudinal spin susceptibility using Dyson equations and LSDA-based DFT methods.
- Built robust data visualization and numerical processing scripts to analyze large-scale simulation outputs for Materials Theory.

FUTF Campaigns Platform — April 2025
Web Development — *Vue, JavaScript*

- Developed a responsive web platform for Engineering Physics Guild Campaign Site.
- Integrated dynamic frontend components and version control workflows using Git.

DataComparisonBio — Febraury 2024 – April 2024
Bioinformatics Analysis Project — *Python*

- Created Python tools to compare genomic datasets and extract statistical differences across samples.
- Implemented visualization pipelines and automated data comparison routines to enhance reproducibility.

RoboticHandControl — August 2023 – June 2024
Embedded Systems Project — *C++, Arduino*

Designed and programmed a remote-controlled bionic hand using servo actuation and wireless signal processing.

Awarded “Best Practical Graduation Project” for demonstrating advanced mechanical and software integration.

GraphCalculator March 2023 – May 2023
Scientific Computing *Python, NumPy, Matplotlib, Tkinter*

Developed an interactive graphing calculator capable of plotting, differentiation, and intersection analysis.

Implemented a GUI and numerical computation features for mathematical visualization.

Student Engagement

Utnarm 2026, UTN March 2026 – November 2026
Committee member and PR - Manager *Uppsala, Sweden*

F1RST – Uppsala, F1RST May 2025 – May 2026
Board Member and Head of Community *Uppsala, Sweden*

UTN Council (FUM), UTN Jan 2026 – December 2026
Convening Representative (Sammankallande Ledamot) *Uppsala, Sweden*

Utnarm 2025, UTN September 2025 – November 2025
Corporate Host (Företagsvärd) *Uppsala, Sweden*

TDB – Reception 2025, UTN May 2025 – September 2025
Head Buddy (Huvudfadder) *Uppsala, Sweden*

Forsränning 2025, UTN April 2025
Pub and Bar Host *Uppsala, Sweden*

Markerting and Information Group, FUTF August 2025 – September 2026
Content Creator and Graphic Designer *Uppsala, Sweden*

Education and Student Enviromment Committee, FUTF August 2024 – September 2026
Course and Safety Representative *Uppsala, Sweden*

IT Group, FUTF August 2024 – September 2025
Developer, Graphic Designer *Uppsala, Sweden*

Café Group, FUTF August 2024 – September 2025
Café Host *Uppsala, Sweden*

Fredagsfika, FUTF August 2024 – September 2025
Baker *Uppsala, Sweden*

Student Ambassador, NTI Gymnasium August 2022 – June 2024
Ambassador and Student representative *Uppsala, Sweden*

Skills

Programming: Python, C++, C#, Arduino, JavaScript, HTML/CSS, MATLAB

Frameworks & Libraries: Vue.js, React, NumPy, SciPy, Matplotlib, Tkinter, Unity, Flutter

Scientific & Computational: Numerical Methods, Data Processing, Simulation Analysis

Software & Design: Git, LaTeX, Figma, Microsoft Office Suite (Excel, PowerPoint, Word), Slack

Languages: English (Fluent), Swedish (Fluent), German (A2), Hindi (A2 – Native spoken)

References

References and letters of recommendation are available upon request.